Solution Requirements

Date	8 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system should allow authorized bank users to log in securely.	High
2	Authorization levels	Admin can manage the system, while credit analysts can enter applicant details and view predictions.	Medium
3	External interfaces	The system should connect with the Flask web interface, trained ML model, and IBM Watson Machine Learning service.	High

4	Transactions processing	The system should accept applicant details, preprocess the data, run the model, and generate an approval or rejection result.	High
5	Reporting	The system should display prediction results, model name, and approval or rejection probability.	Medium
6	Business rules, etc.	Applicants with poor credit history or serious overdue payments should be classified as high-risk.	High
7	Compliance to laws or regulations	Applicant data should be handled securely and used only for prediction and educational purposes.	High
8	*Other*	The system should validate user inputs and prevent incomplete or incorrect form submission.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate predictions quickly after the user submits the form.	Prediction result within 2–3 seconds
2	Scalability	The system should support more users and applicant records when deployed on the cloud.	Supports at least 100 concurrent users
3	Security & Data Privacy	Applicant details should be protected from unauthorized access.	Secure login, HTTPS, and protected data storage
4	Reliability & Availability	The application should provide consistent predictions without frequent failures.	At least 99% availability
5	Usability & Accessibility	The web application should be simple, responsive, and easy to use on mobile and desktop.	Clear form, readable text, responsive design
6	*Other*	The code should be easy to maintain, update, and deploy on different platforms.	Modular code and documented files